

MENU DRIVEN PROGRAM USING SWITCH-CASE STATEMENT

Name	Avinash singh
SAP ID	590033908
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1. Aim

To design and implement a menu-driven C program using the switch-case control statement that inputs a positive integer from the user and, based on the option chosen, either:

- finds the sum and product of the first and last digit of the number, or
- counts how many digits of the number are lesser than, equal to, and greater than a given digit k.

2. Algorithm

Main Menu

1. Start.
2. Declare the required variables (num, choice, k, digit counters, etc.).
3. Input a positive integer num from the user.
4. If num is not positive, display an error message and stop.
5. Display the menu with two options and input the user's choice.
6. Using switch(choice), go to Case 1 or Case 2 as selected.
7. Display the result obtained.
8. Stop.

Case 1 — Sum and Product of First and Last Digit

1. Store the original number in a temporary variable temp.
2. Find the last digit: $\text{lastDigit} = \text{num} \% 10$.
3. Repeatedly divide temp by 10 ($\text{temp} = \text{temp} / 10$) until temp becomes a single digit; that value is the firstDigit.
4. Compute $\text{sum} = \text{firstDigit} + \text{lastDigit}$.
5. Compute $\text{product} = \text{firstDigit} * \text{lastDigit}$.
6. Display sum and product.

Case 2 — Count Digits Lesser, Equal and Greater than k

1. Input the comparison digit k.
2. Initialize $\text{countLess} = 0$, $\text{countEqual} = 0$, $\text{countGreater} = 0$.
3. Repeat while $\text{temp} > 0$:
 - a. Extract the last digit: $\text{digit} = \text{temp} \% 10$.
 - b. If $\text{digit} < k$, increment countLess; else if $\text{digit} == k$, increment countEqual; else increment countGreater.
 - c. Remove the last digit: $\text{temp} = \text{temp} / 10$.
4. Display countLess, countEqual and countGreater.

3. Test Case Table

S.No.	Input Number	Choice	Extra Input (k)	Expected Output	Actual Output	Result
1	12345	1	-	Sum=6, Product=5	Sum=6, Product=5	Pass
2	12345	2	k=3	Less=2, Equal=1, Greater=2	Less=2, Equal=1, Greater=2	Pass
3	7	1	-	Sum=14, Product=49	Sum=14, Product=49	Pass
4	1000	2	k=0	Less=0, Equal=3, Greater=1	Less=0, Equal=3, Greater=1	Pass
5	456	3 (invalid)	-	"Invalid choice!" message	"Invalid choice!" message	Pass
6	-5 (invalid)	-	-	"Invalid input!" message	"Invalid input!" message	Pass

All six test cases (normal, single-digit, boundary k, invalid menu choice, and invalid negative input) were executed and matched the expected output.

4. Source Code (digit_ops.c)

```
/* =====
Program : Menu Driven Program using Switch-Case
Author  : Rudra Pratap Singh
SAP ID  : 590032811
Batch   : 16
Date    : 5 September 2026

Description:
This program accepts a positive integer from the user and
then, using a switch-case menu, performs ONE of the
following operations chosen by the user:

    1. Find the SUM and PRODUCT of the first and last digit
       of the number.
    2. Count how many digits of the number are LESSER THAN,
       EQUAL TO, and GREATER THAN a digit 'k' given by the
       user.
===== */

#include <stdio.h>
#include <math.h>

int main()
{
    long int num, temp;
    int choice;
    int firstDigit, lastDigit, sum, product;
    int k, digit, countLess, countEqual, countGreater;

    printf("=====\n");
    printf(" MENU DRIVEN PROGRAM (DIGIT OPERATIONS)\n");
    printf("=====\n");

    printf("Enter a positive integer: ");
    scanf("%ld", &num);

    /* Validate that the number entered is positive */
    if (num <= 0)
    {
        printf("Invalid input! Please enter a positive integer.\n");
        return 0;
    }

    printf("\n----- MENU ----- \n");
    printf("1. Sum and Product of first and last digit\n");
    printf("2. Count digits lesser, equal and greater than k\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            temp = num;

            /* last digit = remainder when divided by 10 */
            lastDigit = num % 10;

            /* first digit = keep dividing by 10 till one digit is left */
            while (temp >= 10)
```

```

        {
            temp = temp / 10;
        }
        firstDigit = temp;

        sum = firstDigit + lastDigit;
        product = firstDigit * lastDigit;

        printf("\nNumber entered    : %ld\n", num);
        printf("First digit       : %d\n", firstDigit);
        printf("Last digit        : %d\n", lastDigit);
        printf("Sum of digits      : %d\n", sum);
        printf("Product of digits : %d\n", product);
        break;

    case 2:
        printf("Enter the digit k (0-9) to compare with: ");
        scanf("%d", &k);

        countLess = 0;
        countEqual = 0;
        countGreater = 0;
        temp = num;

        /* extract every digit one by one and compare with k */
        while (temp > 0)
        {
            digit = temp % 10;

            if (digit < k)
                countLess++;
            else if (digit == k)
                countEqual++;
            else
                countGreater++;

            temp = temp / 10;
        }

        printf("\nNumber entered : %ld\n", num);
        printf("Value of k       : %d\n", k);
        printf("Count_lesser    : %d\n", countLess);
        printf("Count_equal     : %d\n", countEqual);
        printf("Count_greater   : %d\n", countGreater);
        break;

    default:
        printf("\nInvalid choice! Please enter either 1 or 2.\n");
}

return 0;
}

```

5. Compilation and Execution

The program was compiled and executed using the GCC compiler from the VS Code integrated terminal (PowerShell) as shown below.

Compilation Command

```
PS C:\Users\Rudra Pratap Singh> gcc digit_ops.c -o digit_ops.exe
PS C:\Users\Rudra Pratap Singh>
```

Test Run 1 — Choice 1 (Sum & Product), Number = 12345

```
PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
=====
MENU DRIVEN PROGRAM (DIGIT OPERATIONS)
=====
Enter a positive integer: 12345

----- MENU -----
1. Sum and Product of first and last digit
2. Count digits lesser, equal and greater than k
Enter your choice (1 or 2): 1

Number entered   : 12345
First digit      : 1
Last digit       : 5
Sum of digits    : 6
Product of digits : 5
PS C:\Users\Rudra Pratap Singh>
```

Test Run 2 — Choice 2 (Digit Comparison), Number = 12345, k = 3

```
PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
=====
MENU DRIVEN PROGRAM (DIGIT OPERATIONS)
=====
Enter a positive integer: 12345

----- MENU -----
1. Sum and Product of first and last digit
2. Count digits lesser, equal and greater than k
Enter your choice (1 or 2): 2
Enter the digit k (0-9) to compare with: 3

Number entered : 12345
Value of k      : 3
Count_lesser    : 2
Count_equal     : 1
Count_greater   : 2
PS C:\Users\Rudra Pratap Singh>
```

Test Run 3 — Choice 1, Single Digit Number = 7

```
PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter a positive integer: 7
Enter your choice (1 or 2): 1

Number entered      : 7
First digit         : 7
Last digit          : 7
Sum of digits       : 14
Product of digits   : 49
PS C:\Users\Rudra Pratap Singh>
```

Test Run 4 — Choice 2, Number = 1000, k = 0

```
PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter a positive integer: 1000
Enter your choice (1 or 2): 2
Enter the digit k (0-9) to compare with: 0

Number entered : 1000
Value of k     : 0
Count_lesser   : 0
Count_equal    : 3
Count_greater  : 1
PS C:\Users\Rudra Pratap Singh>
```

Test Run 5 — Invalid Menu Choice

```
PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter a positive integer: 456
Enter your choice (1 or 2): 3

Invalid choice! Please enter either 1 or 2.
PS C:\Users\Rudra Pratap Singh>
```

Test Run 6 — Invalid (Negative) Number

```
PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter a positive integer: -5
Invalid input! Please enter a positive integer.
PS C:\Users\Rudra Pratap Singh>
```